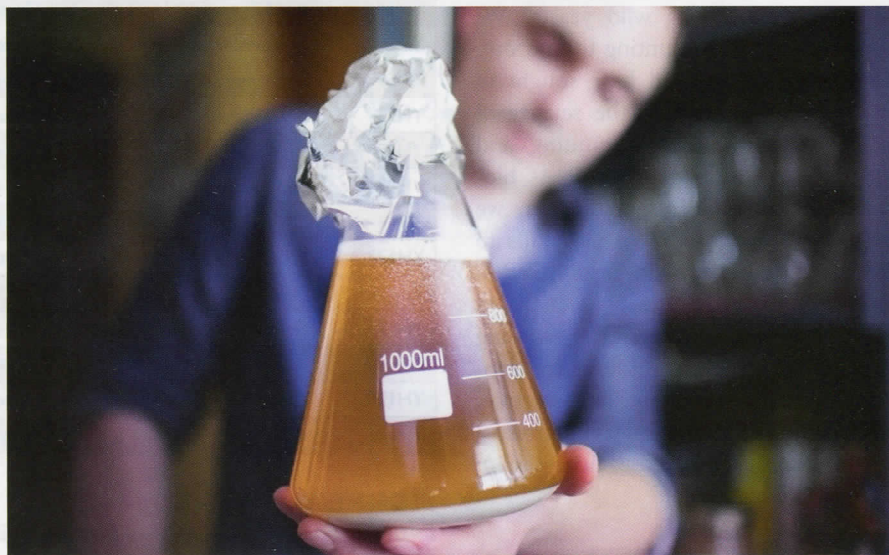


by Kara Osburn, Nur Nisaa Ahmad, and Matthew L. Bochman

The Wild, Wild World of Wild Yeast

Editor's Note: This is the eighth published experiment from the AHA's Research & Education Fund (REF). For more on the REF and to see other completed projects, go to HomebrewersAssociation.org/community/research-and-education-fund. The data and observations presented in this report were derived from only one experiment and therefore should not be considered conclusive.



It has been said that brewers make wort, but yeast makes beer. And it's true: microscopic yeast cells eat wort sugars and create the ethanol, carbon dioxide, and aromatic compounds that comprise our favorite intoxicating, effervescent beverage. That said, yeast tends to be the forgotten ingredient in beer: it was even left out of the original Reinheitsgebot (For more on the Reinheitsgebot, see page 50).

For years, home- and craft brewers have experimented with new hop varieties, vintage and heirloom barley breeds, spices, and even water chemistry, but identification and use of novel yeasts in beer brewing has lagged behind that of these other ingredients.^{1,2,3,4} But in beers that aren't heavily hopped (despite the current IPA craze, such things do exist), yeast can contribute more than half the flavor profile of the finished product. True, there are strains like White Labs WLP001 California Ale Yeast that have neutral flavor profiles and add little more than alcohol and bubbles to your beer. However, there are others that provide the banana

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aromatics (isoamyl acetate) in hefeweizen or the hint of butterscotch (diacetyl) in some English ales, just to name a couple.

As lovers of yeast and the things that yeasts do, my students and I wanted to answer the question, "What other flavors are out there?" There are an estimated 150,000 species of yeast on Earth, but only about one percent of them have been described.⁵ Of these, members of the genus *Saccharomyces*, especially *S. cerevisiae* (ale yeast) and the *S. cerevisiae*/*S. eubayanus* hybrid *S. pastorianus* (lager yeast),

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dominate European and American beer brewing industries.^{6,7} However, pre-industrial brewing often relied upon spontaneous fermentation, which was performed by various yeasts and other microbes native to particular regions.⁸

Have we lost classic beer aromas and flavors by limiting ourselves to species of *Saccharomyces*? What other yeast-borne tastes and smells are waiting to be discovered among wild yeasts? We decided to go yeast hunting to find out. The research presented here sought to bridge the gap between the science of isolating wild yeasts (yeast hunting) and their eventual use in brewing. As biochemists and molecular biologists, we aimed to leverage our wet-lab skills to develop an effective wild yeast isolation and selection procedure, as well as to identify species with desirable brewing characteristics: ethanol tolerance; evolutionary relatedness to *Saccharomyces cerevisiae*; production of pleasant aromatics; metabolism of maltose; flocculation; attenuation; and flavor.

For the sake of brevity, we have intentionally omitted a detailed discussion of our experimental methods. But citizen scientists interested in the full details of our approach are encouraged to consult our AHA REF write-up, available online at HomebrewersAssociation.org.

Rich sources of ethanol-tolerant wild yeasts

In his AHA REF project, Lentz (2014) isolated multiple yeasts from fruits and barley using malt yeast peptone glucose (MYPG) broth, a rich medium that permits the growth of essentially all aerobic yeast and bacteria. After purifying the yeasts of all other organisms, he tested them for ethanol tolerance, which is a necessary attribute of any beer yeast.

We used a different tactic for our experiment by employing a rich medium containing 5 percent ethanol to immediately isolate only ethanol-tolerant yeasts. The average craft beer contains less than 6 percent alcohol by volume, so using this selection, we ensured that every organism that we isolated could at least survive in an ethanol concentration commonly

TABLE 1:
Various Isolated Yeasts, Their Sources, and Their Aroma Profiles When Grown In Liquid Culture

Strain	Genus and species	Source	Aroma
YH1	<i>Hanseniospora uvarum</i>	Serviceberry	Mild rubbing alcohol
YH2	<i>Hanseniospora uvarum</i>	Serviceberry	Yeasty
YH5	<i>Starmerella bombicola</i>	Dandelion blossom	Neutral to mildly musty
YH6	<i>Cryptococcus albidus</i>	Dandelion blossom	Yeasty
YH9	<i>Metchnikowia fructicola</i>	Serviceberry	Maple syrup
YH14	<i>Candida carpophila</i>	Dogwood bark	Feet
YH18	<i>Lancea kluyveri</i>	Honey locust bark	Neutral to mild floral
YH19	<i>Torulaspora delbrueckii</i>	Weeping cherry bark	Strong floral, nearly astringent
YH23	<i>Kluyveromyces lactis</i>	Honey locust bark	Mild floral
YH25	<i>Lancea fermentati</i>	Red oak bark	Floral
YH35	<i>Saccharomyces paradoxus</i>	Red oak bark	Neutral
YH44	<i>Saccharomyces cerevisiae</i>	Shagbark hickory bark	Yeasty
YH123	<i>Saccharomyces cerevisiae</i>	White oak bark	Sweet, floral
YH138	<i>Debaromyces hansenii</i>	Soil	Light floral
YH156	<i>Schizosaccharomyces japonicas</i>	Oak bark	Mesquite smoke
YH169	<i>Brettanomyces bruxellensis</i>	Wild fermented beer	Neutral
YH178	<i>Torulaspora delbrueckii</i>	Honey bee	Fruity/citrus
YH196	<i>Saccharomyces cerevisiae</i>	Pawpaw fruit	Earthy

FIGURE 1:
Evolutionary Relationships among Wild Strains and Four Commercially Available Brewer's Yeasts

(A) Representative DNA sequences from the indicated strains were aligned, and the evolutionary relationships among them were determined. The closer two strains cluster on the tree, the closer their relationship. The commercially available strains are from White Labs (WLP001, WLP300, and WLP715) and Wyeast (WY2007). (B) Enlarged and annotated "Saccharomyces species" portion of the tree from (A). The genus abbreviations S., T., and W. stand for *Saccharomyces*, *Torulaspora*, and *Wickerhamomyces*, respectively.

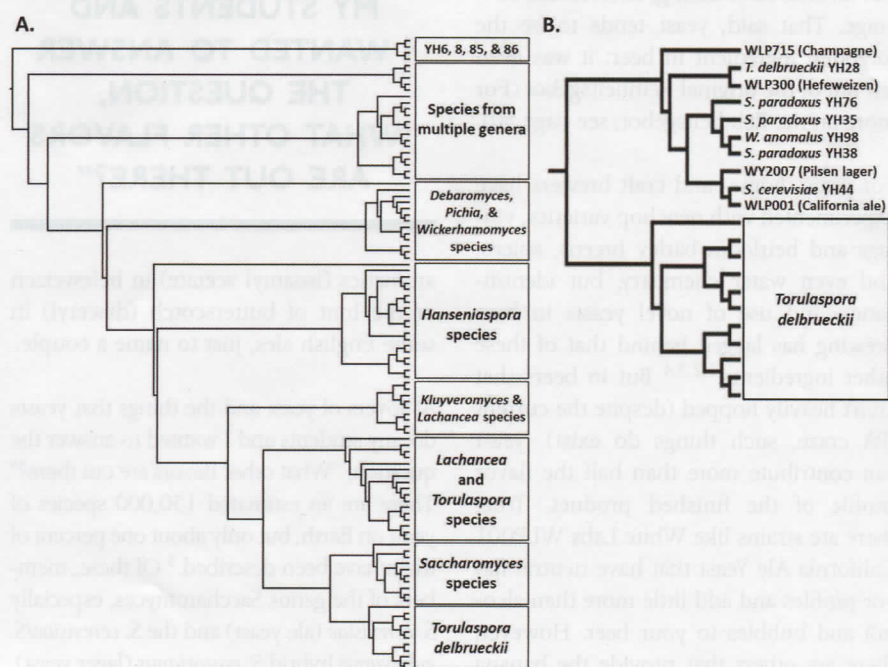
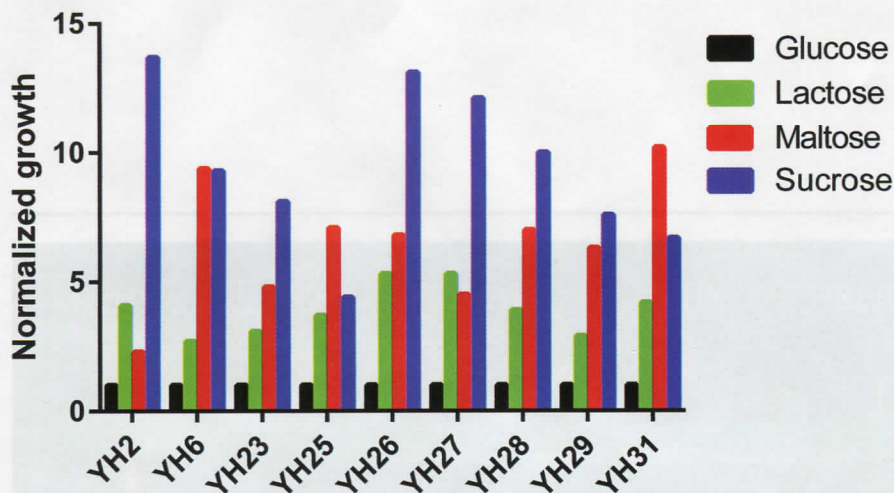
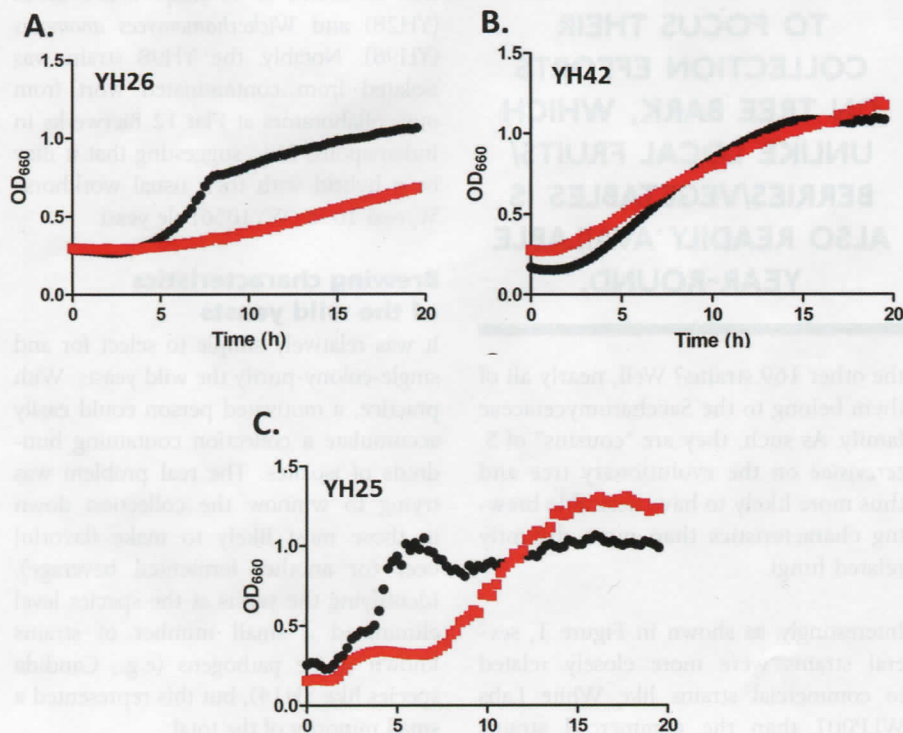


FIGURE 2:**Relative Yeast Growth in the Presence Of Various Sugars**

The indicated yeast strains were grown in a rich medium containing a single sugar (glucose, lactose, maltose, or sucrose), and their growth was normalized to that in a glucose-containing medium. Values greater than one indicate better growth.

**FIGURE 3:****Growth Kinetics of Strains in Glucose- and Maltose-Containing Media**

The indicated strains were grown overnight in media containing either glucose (black) or maltose (red). Growth was then monitored for about 20 hours and is plotted as the optical density at 660 nm (OD_{660}) vs. time.



HAVE WE LOST CLASSIC BEER AROMAS AND FLAVORS BY LIMITING OURSELVES TO SPECIES OF SACCHAROMYCES?

found in beer.⁹ Including ethanol in the selection medium also eliminated the growth of nearly all other organisms (i.e., bacteria and filamentous fungi), which generally have lower alcohol tolerances than *S. cerevisiae* and related yeasts.

Like bacteria, yeasts are ubiquitous in nearly all environments. Unfortunately, relatively few are suitable for beer brewing, making yeast hunting a true “needle in the haystack” proposition. One might naively assume that natural sources of concentrated sugar, such as fruits and berries, would be the best sources of brew-worthy yeasts. Indeed, an earlier AHA Research and Education Fund (REF) project concerned the isolation and initial characterization of wild yeasts and largely focused on fruit.⁴ However, levels of *S. cerevisiae* and related brewing species on such substrates are much lower than those of other yeasts: just one of every 1,000 grapes harbors *S. cerevisiae*, and those that do carry it host 100 to 1,000 times as many cells from other yeast species.¹⁰ Instead, research demonstrates that tree bark, especially from oak, is a natural reservoir of *Saccharomyces* species.^{11,12}

We began yeast hunting locally in Bloomington, Ind. to isolate and characterize new strains for brewing but also extended our search to nearby states (West Virginia, Pennsylvania, Ohio, New Jersey, and New York). Because yeasts are found everywhere from the Antarctic to the ocean floor, we took an unbiased approach when screening environmental samples for yeast.^{13,14} Table 1 shows a random assortment of strains that we identified and the sources from which they were isolated.

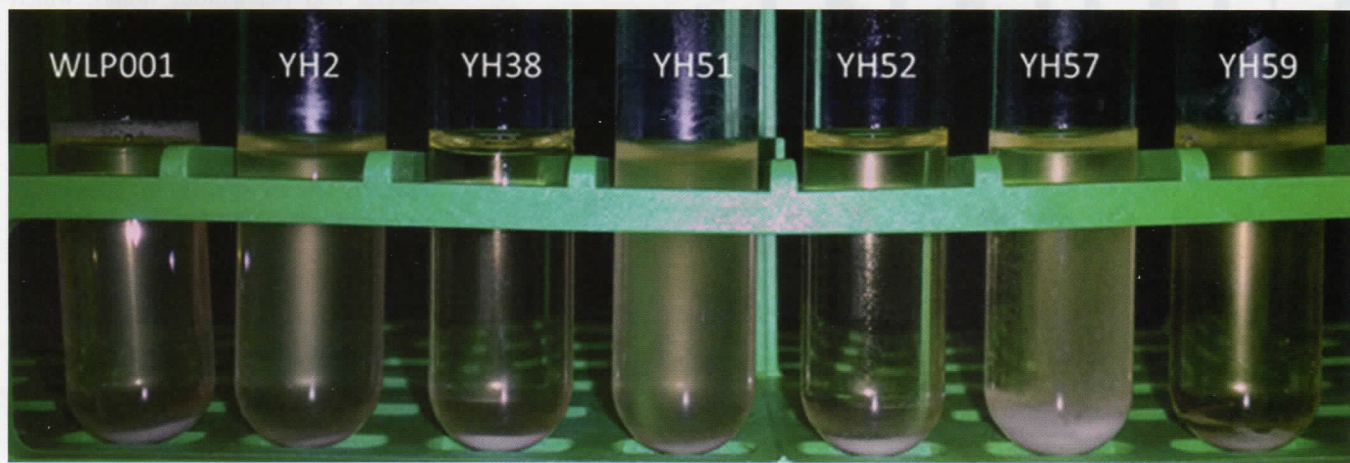
During this ongoing experiment, we have successfully isolated 200 different strains

FIGURE 4:

Qualitative Flocculation Assessment

The indicated strains were inoculated into rich medium, incubated at 86° F (30° C) with aeration for four hours, and then incubated without aeration (i.e., vertically in a test tube rack) at 86° F (30° C) for two days. Good flocculation was evident by the clarity of the culture, with strains less able to flocculate resulting in increasingly turbid media. Flocculation behavior from best to worst was as follows:

1. YH52
2. YH38
3. WLP001 and YH59
4. YH2
5. YH57
6. YH51



from multiple types of tree bark, soil, flowers, fruits, berries, vegetables, and contaminated beer (Table 1). Of these, nearly two thirds were isolated from tree bark. Therefore, based on our success rate of the number of ethanol-tolerant yeasts isolated per sample, we would advise other yeast hunters to focus their collection efforts on tree bark, which unlike local fruits/berries/vegetables is also readily available year-round. While we did find potentially brew-worthy yeasts on all types of samples, repeated sampling was sometimes necessary to hit pay dirt.

Diversity of yeasts identified

Our collection of 200 wild yeasts features multiple strains from more than 30 species. This includes 14 wild *S. cerevisiae* strains, eight isolates of *Saccharomyces paradoxus* (the closest known relative of *S. cerevisiae*), three *Brettanomyces bruxellensis* strains, and six species commonly associated with African millet beer production, *Schizosaccharomyces pombe*. What about

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the other 169 strains? Well, nearly all of them belong to the Saccharomycetaceae family. As such, they are “cousins” of *S. cerevisiae* on the evolutionary tree and thus more likely to have desirable brewing characteristics than more distantly related fungi.

Interestingly, as shown in Figure 1, several strains were more closely related to commercial strains like White Labs WLP001 than the commercial strains

were to each other. Unsurprisingly, this included the wild *S. cerevisiae* and *S. paradoxus* strains mentioned above, as well as strains of *Torulaspora delbrueckii* (YH28) and *Wickerhamomyces anomalus* (YH98). Notably, the YH98 strain was isolated from contaminated wort from our collaborators at Flat 12 Bierwerks in Indianapolis, Ind., suggesting that it may be a hybrid with their usual workhorse Wyeast 1056 (WY1056) ale yeast.

Brewing characteristics of the wild yeasts

It was relatively simple to select for and single-colony-purify the wild yeasts. With practice, a motivated person could easily accumulate a collection containing hundreds of isolates. The real problem was trying to winnow the collection down to those most likely to make flavorful beer (or another fermented beverage). Identifying the yeasts at the species level eliminated a small number of strains known to be pathogens (e.g., *Candida* species like YH14), but this represented a small minority of the total.

TABLE 2:
Normalized Growth of Selected Strains in Media Containing Various Concentrations of Ethanol

Strain	No ethanol	5% ABV	10% ABV	15% ABV
YH2	1.0	0.646 ●	0.479 ●	0.090 ●
YH5	1.0	0.954 ●	0.938 ●	0.246 ●
YH6	1.0	0.300 ●	0.200 ●	0.171 ●
YH7	1.0	0.909 ●	0.273 ●	0.227 ●
YH9	1.0	1.288 ●	0.373 ●	0.254 ●
YH12	1.0	0.831 ●	0.197 ●	0.211 ●
YH15	1.0	0.625 ●	0.798 ●	0.269 ●
YH16	1.0	0.617 ●	0.798 ●	0.670 ●
YH17	1.0	0.838 ●	0.959 ●	0.378 ●
YH18	1.0	0.558 ●	0.690 ●	0.407 ●
YH21	1.0	1.000 ●	1.000 ●	0.900 ●
YH23	1.0	1.011 ●	0.705 ●	0.682 ●
YH24 ●	1.0	1.551 ●	1.551 ●	1.449 ●
YH25	1.0	0.635 ●	0.702 ●	0.721 ●
YH26	1.0	0.405 ●	0.460 ●	0.423 ●
YH27	1.0	0.519 ●	0.549 ●	0.451 ●
YH28	1.0	0.554 ●	0.574 ●	0.541 ●
YH29	1.0	1.180 ●	1.164 ●	0.689 ●
YH30	1.0	0.765 ●	0.696 ●	0.225 ●
YH31	1.0	0.832 ●	0.802 ●	0.822 ●

● Poor ethanol tolerance
 ● Medium ethanol tolerance
 ● High ethanol tolerance
 ● Ethanol promotes growth
 ● Extremely ethanol tolerant

A better culling point was to first grow the isolated strains in liquid media and subject them to the “sniff test.” In other words, we simply smelled the cultures as a primitive but effective first-pass sensory analysis. The growth medium we used had a neutral odor profile, like a blonde wort with low IBUs, so anything above background was due to aromatics produced by the yeast cells. As shown in Table 1, some strains produced astringent odors reminiscent of rubbing alcohol (e.g., YH1) or sweaty feet (YH14). These were eliminated from further testing.

We next measured the ability of the yeast to grow in different sugars. The most concentrated sugar in wort is maltose, so minimally we assayed for good growth in maltose-containing media.¹⁵ However, wort also typically contains high concentrations of sucrose, fructose, glucose, and more complex sugars, in addition to adjuncts in specialty beers like lactose in milk stout. Therefore, we tested yeast growth in the presence of a variety of sugars (see Figure 2 for examples). All strains grew well in the presence of maltose, with some displaying large preferences for maltose over glucose. For example, YH6 and YH31 grew about 10-fold better in the presence of maltose compared to glucose (Figure 2), suggesting that they would happily ferment your wort.

Ideally, a brewing strain also rapidly converts maltose and other wort sugars to ethanol and carbon dioxide. The faster that the “food” in the wort is consumed, the faster the pH drops due to yeast metabolism, and the faster ethanol is produced, the less chance there is for a contaminating organism to ruin a fermentation. Thus, we also measured the kinetics of yeast growth in medium containing glucose vs. maltose (Figure 3).

Many strains behaved similarly to YH26 (Figure 3A), which grew well in the presence of both glucose and maltose but displayed a long lag time in the maltose medium. This indicates that such strains would be slow fermenters. However, other strains like YH42 were equivalently vigor-

TABLE 3:
Fermentation and Sensory Analyses

Strain	Original gravity	Final gravity	Attenuation	Lag time	Sensory notes
Wyeast 1056	1.040 (10.0° P)	1.006 (1.5° P)	85%	12 h	Clean, neutral flavor
Wyeast 1056	1.055 (13.5° P)	1.011 (2.7° P)	80%	12 h	Clean, neutral flavor
YH2	1.040 (10.0° P)	1.022 (5.5° P)	45%	36 h	Tart, spicy flavor
YH38	1.040 (10.0° P)	1.024 (6° P)	40%	≤9 h	Vegetal, dirty
YH39	1.055 (13.5° P)	1.044 (11° P)	19%	12 h	Belgian- & saison-like, spicy
YH44	1.055 (13.5° P)	1.036 (9° P)	33%	12 h	Clean, neutral flavor
YH45	1.055 (13.5° P)	1.044 (11° P)	19%	12 h	Neutral
YH51	1.040 (10.0° P)	1.024 (6° P)	40%	24 h	Tart
YH52	1.040 (10.0° P)	1.014 (3.5° P)	63%	12 h	Belgian phenolic, bubblegum
YH52	1.055 (13.5° P)	1.044 (11° P)	19%	12 h	Belgian-like, spicy
YH57	1.040 (10.0° P)	1.016 (4° P)	58%	≤9 h	Strong diacetyl, rounded
YH59	1.040 (10.0° P)	1.012 (3° P)	68%	24 h	Fairly clean, slightly tart
YH59	1.055 (13.5° P)	1.040 (10° P)	26%	12 h	English-like, malty, spicy
YH79	1.055 (13.5° P)	1.024 (6° P)	65%	12 h	Sour, soft, clean

ous in both glucose and maltose medium (Figure 3B). This is essentially the same growth pattern displayed by commercially available strains like WLP001 (data not shown), suggesting that these yeasts would quickly ferment wort to outcompete spoilage organisms.

Notably, these kinetic experiments were an important complement to those used to generate the data in Table 2, which only measured the endpoint. For instance, one might assume that strains such as YH6, 25, and 31, which grew to 9.4-, 7.1-, and 10.2-fold higher

density, respectively, in maltose medium compared to glucose, would perform well during fermentation. However, as shown in Figure 3C for YH25, this strain does grow to higher density in the presence of maltose but displays a much longer lag time during which spoilage could occur.

Next, flocculation was assessed by growing the cultures in stationary tubes, and qualitatively judged by visually inspecting the turbidity of the cultures (Figure 4). Compared to WLP001, we isolated strains with better flocculation characteristics

(e.g., YH38 and YH52), similar levels of flocculation (e.g., YH59), and less ability to flocculate (e.g., YH2, YH51, and YH57, which were on par with hefeweizen yeast flocculation).

Ethanol tolerance of the wild yeasts

As stated earlier, the average craft beer contains less than 6 percent ABV (Nelson 2014), and all of the strains in our collection can tolerate at least 5 percent ABV. Craft and homebrewers are known for pushing boundaries, though, so we also tested growth of our strains in media containing higher concentrations of ethanol and compared that to growth in media lacking ethanol (representative data shown in Table 2).

We found that some strains like YH6 were barely ethanol tolerant, with even 5 percent ABV decreasing growth by 70 percent. However, most strains displayed at most a 50 percent decrease in growth in the presence of 5 percent ABV, and many similarly tolerated 10 percent ABV (e.g., YH30). Some strains like YH9 and YH29 even grew better in the presence of low concentrations of ethanol compared to media without ethanol. Rare isolates such as YH24, a strain of *Kluyveromyces marxianus*, not only tolerated up to 15 percent ABV but grew better than in media lacking ethanol altogether. We conclude that one could modify our selection procedure to specifically hunt for extremely ethanol-tolerant yeasts suitable for high-gravity fermentation.

Trial brewing

“The proof of the pudding is in the eating,” and the usefulness of the yeast is only fully obvious in the brewing. After collecting species data and determining a handful of brewing characteristics in the lab, we triaged the initial collection of 100 down to the 10 best candidates and put those yeasts to the ultimate test by brewing with them. We did this in collaboration with the mad scientist brewers at Flat 12 Bierwerks in Indianapolis, Ind. To summarize the process, we started from single colonies of the yeast indicated in Table 3 and propagated them in liquid culture to generate 100 billion cells, which Flat

12 then pitched into 5-gallon (18.93 L) batches of wort. As a control, we cultured and pitched Wyeast 1056 in the same manner.

Two rounds of these brewing experiments were performed using porter wort with an original gravity (OG) of 1.040 (10° P) in September 2014 and 1.055 (13.5° P) in January 2015 (Table 3, white and shaded, respectively). In two cases (YH52 and YH59, *T. delbrueckii* and *Kluyveromyces lactis*), strains that performed well in the first experiment were tested again in the second. In both cases, the sensory analysis was performed by BJCP-certified judges at the brewery. Public tastings were also held at Flat 12 Bierwerks during an event called Defining “Drink Local” (indybeernews.com/blog/defining-drink-local/) and at the 7th Annual Brewers of Indiana Guild Winterfest. Anecdotal reports indicate that the public generally agreed with the BJCP judges concerning the flavors and aromas of the beers, though the average drinker tended to like the sweetness of the poorly attenuated beers better than did their professional counterparts.

As can be seen in Table 3, none of the experimental strains performed overall as well as WY1056, which had a short lag time, attenuated well, and produced a clean, neutral flavor profile. However, the *S. paradoxus* strain YH38 and *Lachancea thermotolerans* strain YH57 displayed a considerably shorter lag time than WY1056, three strains yielded at least 63 percent attenuation (*T. delbrueckii* YH52, *K. lactis* YH59, and *L. thermotolerans* YH79), and many resulted in neutral or pleasing flavor profiles. This suggests that repeated brewing with one or more of these strains could adapt them to better performance in wort. Central State Brewing in Indianapolis took this approach with their house *Brettanomyces bruxellensis* strain, which mutated over a year of culturing and brewing into a quick-attenuating yeast with an excellent flavor profile that can be tweaked depending on the beer style and fermentation conditions used (see centralstatebrewing.com/blog/ for more details).

Overall, the wild *Saccharomyces* strains YH38 and YH44 produced perhaps the least interesting beers. The general favorite and biggest surprise was *L. thermotolerans* YH79, which attenuated well compared to the other wild strains, had a lag time similar to WY1056, and actually soured the porter. Those familiar with sour beers like Belgian lambics and American wild ales know that souring is generally due to the combination of many organisms, including multiple species of yeast and *Lactobacillus* bacteria.¹⁶ Having a pure culture of a single yeast species that can sour beer is easier than trying to wrangle multiple microbes, and we are following up on YH79 in ongoing experiments.

Conclusions

Did we find a million-dollar yeast? Well, let's just say that White Labs hasn't come begging to be our distributor yet. That said, we were able to develop a simple protocol to isolate ethanol-tolerant yeasts from rich natural reservoirs, cull the herd down to the best brewing prospects, and make drinkable (and often delicious) beer. This was somewhat easier for us to do in a lab setting than it would be for the average homebrewer or even craft brewer, but homebrewers are all scientists themselves.

The performance characteristics of our favored yeast strains might be improved through selective isolation in successive generations. Strains of brewer's yeast are ultimately chosen from those cells that efficiently ferment wort and create appealing aromas and flavors along the way. Wild yeasts are more likely to do this when they are isolated from malt-based media since yeasts adapt to their preferred substrates. As this process is repeated, it's possible that some of our strains will improve beyond what we observed in this first series of experiments.

We hope that others are inspired to find their own ultra-local wild strains to brew with, and we're going to continue to characterize the isolates in our collection. Prost!

Resources

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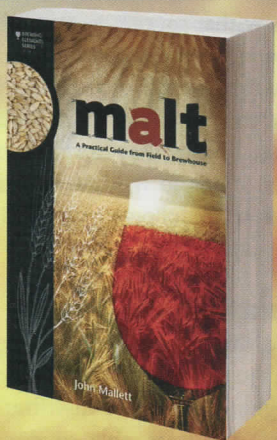
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Kara Osburn, Nur Nisaa Ahmad, and Matthew L. Bochman are members of the Bochman Lab, part of the Department of Molecular and Cellular Biochemistry at Indiana University in Bloomington, Ind. They wish to thank members of the Bochman and van Kessel labs for insight and comments; Rob Caputo, Sean Manahan, and Sean O'Connor for brewing expertise; the AHA REF, the Center of Excellence for Women in Technology's Research Experiences for Undergraduates (CEWiT-REU) program, and the Robert and Marjorie Mann Summer scholarships for funding; and the many volunteers who provided yeast-hunting samples. 🍷



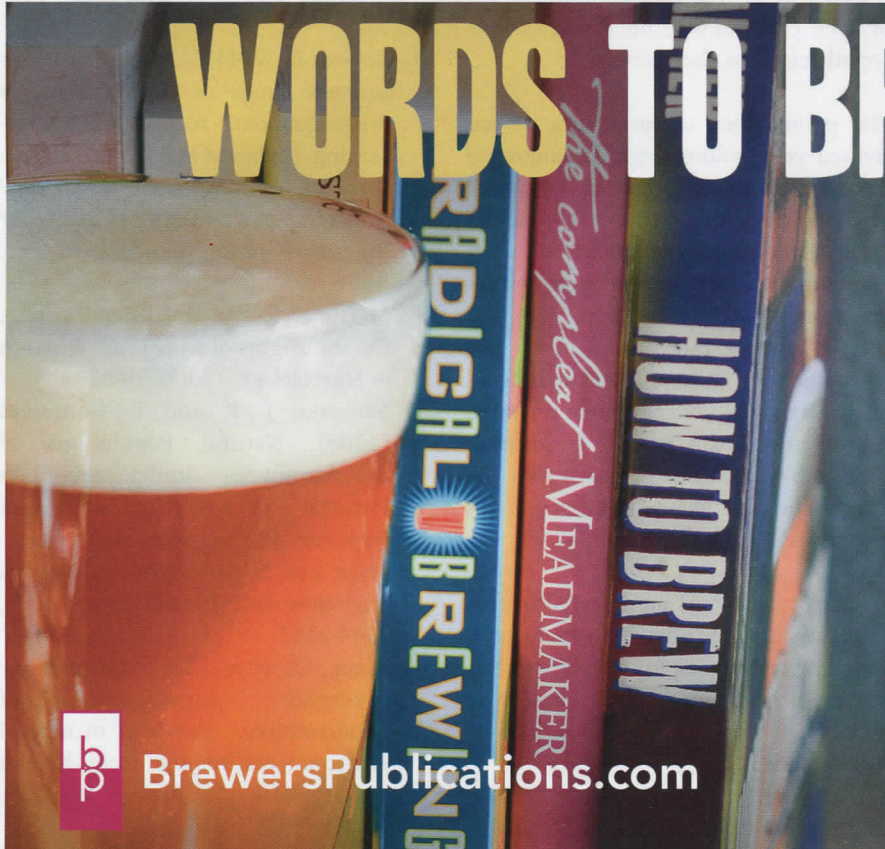
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